

Clustering Algorithms

K-Means (1D & 2D) | Hierarchical Clustering | Dendrogram Construction

Week 14 — Machine Learning

1. What is Clustering?

Clustering is a type of unsupervised machine learning where the goal is to group similar data points together without using any pre-defined labels. The machine must discover the natural structure hidden inside the data entirely on its own.

Real-life analogy: Imagine you tip a bag of mixed sweets onto a table. Without anyone telling you the categories, you naturally sort them by colour, shape, or flavour. That process — finding groups without instructions — is exactly what clustering algorithms do.

A cluster is a collection of data points that are more similar to each other than they are to points in other clusters. Similarity is measured using a distance formula:

- Euclidean Distance (straight-line): $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Manhattan Distance (grid-based): $d = |x_2 - x_1| + |y_2 - y_1|$

Important note: For 1-dimensional data (single feature), Manhattan and Euclidean distance are identical — both equal $|a - b|$.

2. K-Means Clustering — 1-Dimensional Example

2.1 Concept and Algorithm

K-Means is the most widely used clustering algorithm. It represents each cluster by its centroid (centre point) and assigns every data point to its nearest centroid. The algorithm repeats until the centroids stop moving.

Core idea: Pick K locations as starting centres. Every point joins its nearest centre. Recalculate each centre as the mean of its members. Repeat until stable.

The 4 Steps of K-Means

Step 1 Choose K random data points as initial centroids

Step 2 Assign each data point to its closest centroid (using distance)

Step 3 Recalculate each centroid = mean of all points assigned to it

Step 4 Repeat Steps 2 and 3 until centroids do not change (convergence)

2.2 Worked Example — 1D Data, K=2

Dataset: Data = { 2, 4, 10, 12, 3, 11 } K = 2 clusters Distance = Manhattan

Step 1 Choose initial centroids

We randomly select two data points as our starting centroids:

$C1 = 2$ $C2 = 10$

These are just random guesses. The algorithm will move them to better positions through iteration.

Step 2**Assign each point to its nearest centroid (Iteration 1)**

For each data point, calculate its Manhattan distance to $C1=2$ and $C2=10$. Assign the point to whichever centroid is closer.

Data Point	$ \text{point} - C1=2 $	$ \text{point} - C2=10 $	Nearest	Assign to
2	$ 2-2 = 0$	$ 2-10 = 8$	$0 < 8$	C1
4	$ 4-2 = 2$	$ 4-10 = 6$	$2 < 6$	C1
10	$ 10-2 = 8$	$ 10-10 = 0$	$0 < 8$	C2
12	$ 12-2 = 10$	$ 12-10 = 2$	$2 < 10$	C2
3	$ 3-2 = 1$	$ 3-10 = 7$	$1 < 7$	C1
11	$ 11-2 = 9$	$ 11-10 = 1$	$1 < 9$	C2

After Iteration 1: Cluster 1 = {2, 4, 3} Cluster 2 = {10, 12, 11}

Step 3**Recalculate centroids**

The new centroid of each cluster is the arithmetic mean of all points in that cluster.

$$\text{New } C1 = \text{mean}(\{2, 4, 3\}) = (2 + 4 + 3) / 3 = 9 / 3 = 3$$

$$\text{New } C2 = \text{mean}(\{10, 12, 11\}) = (10 + 12 + 11) / 3 = 33 / 3 = 11$$

Centroids have moved: C1 shifted from 2 to 3, and C2 shifted from 10 to 11. We must now check whether any assignments change with these new centres.

Step 2**Reassign all points with new centroids $C1=3$, $C2=11$ (Iteration 2)**

Data Point	$ \text{point} - C1=3 $	$ \text{point} - C2=11 $	Nearest	Assign to
2	$ 2-3 = 1$	$ 2-11 = 9$	$1 < 9$	C1
4	$ 4-3 = 1$	$ 4-11 = 7$	$1 < 7$	C1
10	$ 10-3 = 7$	$ 10-11 = 1$	$1 < 7$	C2

Data Point	$ \text{point} - C1=3 $	$ \text{point} - C2=11 $	Nearest	Assign to
12	$ 12-3 = 9$	$ 12-11 = 1$	$1 < 9$	C2
3	$ 3-3 = 0$	$ 3-11 = 8$	$0 < 8$	C1
11	$ 11-3 = 8$	$ 11-11 = 0$	$0 < 8$	C2

The cluster memberships are IDENTICAL to Iteration 1. This means the centroids will also remain at $C1=3$ and $C2=11$.

Convergence: When cluster assignments do not change between two consecutive iterations, the algorithm has converged. We stop here.

FINAL ANSWER: Cluster 1 = {2, 3, 4} (centroid = 3) Cluster 2 = {10, 11, 12} (centroid = 11)

Interpretation: The algorithm automatically discovered that 2, 3, and 4 are naturally grouped together (small values) and 10, 11, 12 are another natural group (large values). The boundary falls in the gap between 4 and 10.

3. K-Means Clustering — 2-Dimensional Example

3.1 Why 2D is Different

In 2D (or higher dimensions), each data point has two features — for example, a student's height and weight, or a customer's age and spending. We now use Euclidean distance to measure how far apart two points are in this 2D space.

$$\text{Euclidean Distance} = \text{sqrt}((x_2 - x_1)^2 + (y_2 - y_1)^2)$$

The centroid in 2D is calculated separately for each dimension: $\text{new_x} = \text{mean of all x-values in cluster}$, $\text{new_y} = \text{mean of all y-values in cluster}$.

3.2 Worked Example — 2D Data, K=2

Dataset: 6 data points with 2 features each (x, y). Think of these as 6 shops with location coordinates. K = 2 clusters. Distance = Euclidean.

Point	x (feature 1)	y (feature 2)
A	1	1
B	2	1
C	4	3
D	5	4
E	1	2
F	4	4

Step 1 Choose K=2 random initial centroids

We randomly choose Point A and Point D as our starting centroids:

$$C1 = A = (1, 1) \qquad C2 = D = (5, 4)$$

Step 2 Assign each point to its nearest centroid — Iteration 1

For each of the 6 points, calculate Euclidean distance to $C1=(1,1)$ and $C2=(5,4)$:

Distance calculations:

Point A = (1,1):

dist to C1(1,1) = $\sqrt{(1-1)^2 + (1-1)^2}$ = $\sqrt{0+0}$ = 0.00

dist to C2(5,4) = $\sqrt{(1-5)^2 + (1-4)^2}$ = $\sqrt{16+9}$ = 5.00

--> Assign to C1 (0.00 < 5.00)

Point B = (2,1):

dist to C1(1,1) = $\sqrt{(2-1)^2 + (1-1)^2}$ = $\sqrt{1+0}$ = 1.00

dist to C2(5,4) = $\sqrt{(2-5)^2 + (1-4)^2}$ = $\sqrt{9+9}$ = 4.24

--> Assign to C1 (1.00 < 4.24)

Point C = (4,3):

dist to C1(1,1) = $\sqrt{(4-1)^2 + (3-1)^2}$ = $\sqrt{9+4}$ = 3.61

dist to C2(5,4) = $\sqrt{(4-5)^2 + (3-4)^2}$ = $\sqrt{1+1}$ = 1.41

--> Assign to C2 (1.41 < 3.61)

Point D = (5,4):

dist to C1(1,1) = $\sqrt{(5-1)^2 + (4-1)^2}$ = $\sqrt{16+9}$ = 5.00

dist to C2(5,4) = $\sqrt{(5-5)^2 + (4-4)^2}$ = $\sqrt{0+0}$ = 0.00

--> Assign to C2 (0.00 < 5.00)

Point E = (1,2):

dist to C1(1,1) = $\sqrt{(1-1)^2 + (2-1)^2}$ = $\sqrt{0+1}$ = 1.00

dist to C2(5,4) = $\sqrt{(1-5)^2 + (2-4)^2}$ = $\sqrt{16+4}$ = 4.47

--> Assign to C1 (1.00 < 4.47)

Point F = (4,4):

dist to C1(1,1) = $\sqrt{(4-1)^2 + (4-1)^2}$ = $\sqrt{9+9}$ = 4.24

dist to C2(5,4) = $\sqrt{(4-5)^2 + (4-4)^2}$ = $\sqrt{1+0}$ = 1.00

--> Assign to C2 (1.00 < 4.24)

Point	Coords	Dist to C1(1,1)	Dist to C2(5,4)	Assigned to
A	(1,1)	0.00	5.00	C1
B	(2,1)	1.00	4.24	C1
C	(4,3)	3.61	1.41	C2
D	(5,4)	5.00	0.00	C2
E	(1,2)	1.00	4.47	C1
F	(4,4)	4.24	1.00	C2

Cluster 1 = {A(1,1), B(2,1), E(1,2)}

Cluster 2 = {C(4,3), D(5,4), F(4,4)}

Step 3

Recalculate 2D centroids

In 2D, the new centroid = (mean of all x-values, mean of all y-values) in each cluster.

$$\text{New C1}_x = (1+2+1)/3 = 4/3 = 1.33 \quad \text{New C1}_y = (1+1+2)/3 = 4/3 = 1.33$$

$$\text{New C1} = (1.33, 1.33)$$

$$\text{New C2}_x = (4+5+4)/3 = 13/3 = 4.33 \quad \text{New C2}_y = (3+4+4)/3 = 11/3 = 3.67$$

$$\text{New C2} = (4.33, 3.67)$$

Step 2

Reassign all points with new centroids — Iteration 2

Recalculate distances using C1=(1.33, 1.33) and C2=(4.33, 3.67):

Point	Dist to C1(1.33,1.33)	Dist to C2(4.33,3.67)	Assigned to
A(1,1)	$\sqrt{0.11+0.11} = 0.47$	$\sqrt{10.89+6.99} = 4.23$	C1
B(2,1)	$\sqrt{0.44+0.11} = 0.75$	$\sqrt{5.33+6.99} = 3.51$	C1
C(4,3)	$\sqrt{7.11+2.78} = 3.14$	$\sqrt{0.11+0.45} = 0.75$	C2
D(5,4)	$\sqrt{13.44+7.11} = 4.53$	$\sqrt{0.45+0.11} = 0.75$	C2
E(1,2)	$\sqrt{0.11+0.44} = 0.75$	$\sqrt{10.89+2.78} = 3.70$	C1
F(4,4)	$\sqrt{7.11+7.11} = 3.77$	$\sqrt{0.11+0.11} = 0.47$	C2

Cluster memberships are identical to Iteration 1. Centroids will not change. The algorithm has converged.

FINAL ANSWER: Cluster 1 = {A(1,1), B(2,1), E(1,2)} centroid=(1.33, 1.33)
Cluster 2 = {C(4,3), D(5,4), F(4,4)} centroid=(4.33, 3.67)

Interpretation: The algorithm found two spatial groups — a 'bottom-left' cluster of nearby points and a 'top-right' cluster of nearby points. In a real scenario, these might represent two delivery zones, two customer segments, or two natural groupings of observations.

4. Hierarchical (Agglomerative) Clustering

4.1 Concept

Hierarchical clustering takes a completely different approach from K-Means. Instead of starting with K fixed clusters, it starts with every single data point as its own cluster, then repeatedly merges the two closest clusters, step by step, until all points belong to one big cluster.

Key advantage over K-Means: You do NOT need to specify K before running the algorithm. After the algorithm finishes, you look at the dendrogram and decide where to cut it — giving you any number of clusters you want, all from a single run.

This is called agglomerative (bottom-up) clustering. Imagine starting with a group of strangers, then letting the most similar pairs form duos, then trios, and so on until everyone is one big group.

4.2 The Algorithm — 5 Steps

Step 1 Assign each of the N data points to its own cluster (N clusters total)

Step 2 Compute the pairwise distance between every pair of clusters

Step 3 Find the two closest clusters and **MERGE** them into one

Step 4 Recompute distances between the new merged cluster and all remaining clusters

Step 5 Repeat Steps 3 and 4 until only 1 cluster remains — then draw the dendrogram

4.3 Inter-Cluster Distance Methods

When two clusters have multiple members, we need a rule for defining the 'distance' between them. There are three common rules:

Method	Rule	Effect
Single-Linkage	Distance = SHORTEST distance between any member of cluster A and any member of cluster B	Creates chain-like, elongated clusters
Complete-Linkage	Distance = LONGEST distance between any member of cluster A and any member of cluster B	Creates compact, spherical clusters
Centroid-Linkage	Distance = Distance between the CENTROID (mean point) of cluster A and centroid of cluster B	Balanced compromise between the two

4.4 Worked Example — Complete Solution

Dataset: $X = \{1, 3, 8, 10, 15\}$ Distance = Manhattan Linkage = Single-Linkage

Step 1

Start: 5 data points = 5 clusters

Initial clusters: $\{1\}$ $\{3\}$ $\{8\}$ $\{10\}$ $\{15\}$

Compute ALL pairwise Manhattan distances:

Cluster Pair	Distance Calculation	Distance
$\{1\}$ and $\{3\}$	$ 1-3 = 2$	2 *minimum*
$\{1\}$ and $\{8\}$	$ 1-8 = 7$	7
$\{1\}$ and $\{10\}$	$ 1-10 = 9$	9
$\{1\}$ and $\{15\}$	$ 1-15 = 14$	14
$\{3\}$ and $\{8\}$	$ 3-8 = 5$	5
$\{3\}$ and $\{10\}$	$ 3-10 = 7$	7
$\{3\}$ and $\{15\}$	$ 3-15 = 12$	12
$\{8\}$ and $\{10\}$	$ 8-10 = 2$	2 *tie*
$\{8\}$ and $\{15\}$	$ 8-15 = 7$	7
$\{10\}$ and $\{15\}$	$ 10-15 = 5$	5

Minimum distance = 2, shared by $\{1,3\}$ and $\{8,10\}$. We take the first one found: MERGE $\{1\}$ and $\{3\}$.

After Merge 1: {1,3} {8} {10} {15}

Merge distance = 2

Step 3

Recompute distances — Single-Linkage rule

Distance between cluster {1,3} and another cluster = minimum distance from any member of {1,3} to any member of the other cluster.

Cluster Pair	Single-Linkage Calculation	Distance
{1,3} and {8}	$\min(1-8 , 3-8) = \min(7, 5) = 5$	5
{1,3} and {10}	$\min(1-10 , 3-10) = \min(9, 7) = 7$	7
{1,3} and {15}	$\min(1-15 , 3-15) = \min(14, 12) = 12$	12
{8} and {10}	$ 8-10 = 2$	2 *minimum*
{8} and {15}	$ 8-15 = 7$	7
{10} and {15}	$ 10-15 = 5$	5

Minimum distance = 2, between {8} and {10}. MERGE {8} and {10}.

After Merge 2: {1,3} {8,10} {15}

Merge distance = 2

Step 3

Recompute distances again

Cluster Pair	Single-Linkage Calculation	Distance
{1,3} and {8,10}	$\min(1-8 , 1-10 , 3-8 , 3-10) = \min(7, 9, 5, 7) = 5$	5 *minimum*
{1,3} and {15}	$\min(1-15 , 3-15) = \min(14, 12) = 12$	12
{8,10} and {15}	$\min(8-15 , 10-15) = \min(7, 5) = 5$	5 *tie*

Minimum = 5, tied between {1,3}→{8,10} and {8,10}→{15}. Take the first: MERGE {1,3} and {8,10}.

After Merge 3: {1,3,8,10} {15}

Merge distance = 5

Step 3	Final distance calculation
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Only 2 clusters remain.

$$\{1, 3, 8, 10\} \text{ and } \{15\}: \min(|1-15|, |3-15|, |8-15|, |10-15|) = \min(14, 12, 7, 5) = 5$$

MERGE {1,3,8,10} and {15}. All points now in one cluster.

After Merge 4: {1, 3, 8, 10, 15}	Merge distance = 5
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4.5 Merge History Summary Table

This table records every merge in order, and is used to draw the dendrogram:

Merge #	Clusters Merged	Resulting Cluster	Distance (Height)
1	{1} + {3}	{1, 3}	2
2	{8} + {10}	{8, 10}	2
3	{1,3} + {8,10}	{1, 3, 8, 10}	5
4	{1,3,8,10} + {15}	{1,3,8,10,15}	5

5. How to Draw a Dendrogram

5.1 What is a Dendrogram?

A dendrogram is a tree diagram that visually represents the sequence of cluster merges in hierarchical clustering. The x-axis shows the data points (or cluster labels), and the y-axis shows the distance (or dissimilarity) at which merges occurred.

Reading rule: The HEIGHT of each horizontal bar tells you HOW SIMILAR the merged groups were. Low height = very similar. High height = merged late because they were quite different.

5.2 Step-by-Step Dendrogram Construction

Using the merge history from our example ($X = \{1, 3, 8, 10, 15\}$):

Step A — Place all data points on the x-axis

Draw a horizontal line at $y=0$. Place each data point as a vertical tick mark below it. Order them in sequence: 1, 3, 8, 10, 15.

Step B — Draw Merge 1: $\{1\} + \{3\}$ at height 2

Draw a short vertical line up from point '1' to height 2. Draw another from '3' to height 2. Connect them with a horizontal bar at $y=2$. This U-shape shows that 1 and 3 were merged at distance 2.

Step C — Draw Merge 2: $\{8\} + \{10\}$ at height 2

Same process: vertical lines up from '8' and '10', horizontal bar connecting them at $y=2$. Two separate U-shapes now exist on the diagram.

Step D — Draw Merge 3: $\{1,3\} + \{8,10\}$ at height 5

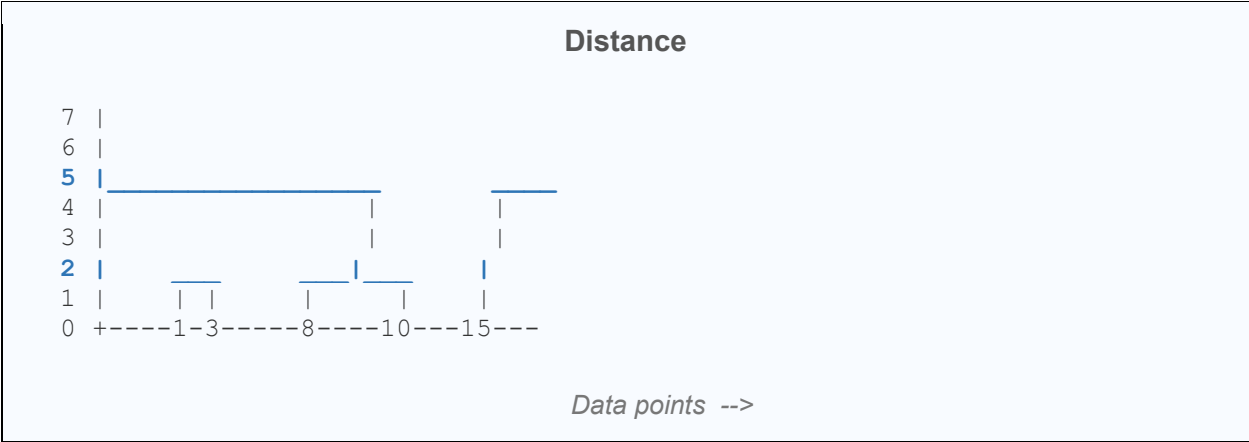
The new cluster $\{1,3\}$ is represented by the midpoint of the bar connecting 1 and 3. The cluster $\{8,10\}$ is represented by its bar's midpoint. Draw vertical lines from each bar UP to height 5, then connect with a horizontal bar at $y=5$. This creates a larger U-shape encompassing 1, 3, 8, and 10.

Step E — Draw Merge 4: $\{1,3,8,10\} + \{15\}$ at height 5

Draw a vertical line up from '15' to height 5. The large cluster {1,3,8,10} is represented at height 5. Connect both to a horizontal bar at y=5 (same height as previous merge — this is valid; they happened to be equidistant).

5.3 Dendrogram Diagram

Below is the dendrogram for $X = \{1, 3, 8, 10, 15\}$:



5.4 How to Extract Clusters from the Dendrogram

The power of a dendrogram is that a single horizontal cut gives you any desired number of clusters. The number of vertical lines the cut crosses equals the number of clusters.

Cut at height	Lines crossed	Number of clusters	Clusters obtained
Above 5 (top)	1	1 cluster	{1, 3, 8, 10, 15}
Between 2 and 5	2	2 clusters	{1,3,8,10} and {15}
Between 2 and 5 (alt cut)	3	3 clusters	{1,3} and {8,10} and {15}
Below 2	5	5 clusters	{1} {3} {8} {10} {15}

Critical insight: With K-Means, if you want to try $K=2$, $K=3$, $K=4$, you must re-run the entire algorithm each time. With hierarchical clustering, you run the algorithm ONCE and simply cut the dendrogram at different heights to get any K you want.

6. Key Differences at a Glance

Property	K-Means	Hierarchical
Starting point	K random centroids chosen upfront	Every point is its own cluster
Requires K upfront?	YES — must decide K before running	NO — decide K after seeing dendrogram
Direction	Top-down: partition data into K groups	Bottom-up: merge pairs until 1 group
Backtracking?	YES — points can move between clusters	NO — once merged, never unmerged
Deterministic?	NO — random init gives different results	YES — same data always gives same result
Unique answer?	NO — depends on random starting points	YES — one definitive dendrogram
Speed	Fast: $O(N \times K \times T)$	Slow: $O(N^2 \log N)$
Works on large data?	YES — scales well	Difficult — memory-heavy for large N
Output	K flat clusters	Full hierarchical dendrogram tree
Sensitivity	Sensitive to initial centroid choice	Sensitive to linkage method chosen
Best used when	Large data, approximate K is known	Small-medium data, exploring structure

The most important difference: K-Means allows reassignment — a data point can switch clusters between iterations as centroids move. Hierarchical clustering is permanent — once two clusters are merged together, that decision is NEVER reversed. This is why hierarchical clustering is called greedy: it makes the locally best decision at each step without reconsidering past choices.

7. Formula Quick Reference

Distance Formulas

Manhattan Distance (1D or nD): $d = |x_1 - x_2| + |y_1 - y_2| + \dots$

Euclidean Distance (2D): $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

K-Means Centroid Update

New centroid = (mean of all x-values in cluster, mean of all y-values in cluster)

For 1D: $\text{new_c} = (x_1 + x_2 + \dots + x_n) / n$

For 2D: $\text{new_c} = ((x_1 + x_2 + \dots + x_n) / n , (y_1 + y_2 + \dots + y_n) / n)$

Davies-Bouldin Index (Clustering Quality)

The DB index measures clustering quality. Lower value = better clustering.

$R_{ij} = [\text{var}(C_i) + \text{var}(C_j)] / || c_i - c_j ||$

$R_i = \max \text{ over all } j \text{ of } R_{ij}$

$DB = (1/K) * \text{sum of all } R_i$

Variance: $\text{var}(x) = (1/(N-1)) * \text{sum of } (x_i - \text{mean})^2$

Hierarchical Linkage Rules

Single-Linkage: $\text{dist}(A,B) = \min \text{ distance between any point in A and any point in B}$

Complete-Linkage: $\text{dist}(A,B) = \max \text{ distance between any point in A and any point in B}$

Centroid-Linkage: $\text{dist}(A,B) = \text{distance between centroid of A and centroid of B}$